Introduction to Quantum Database Principle

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Different Models, the Same Classical Bits

Consider a collection of research papers. What do we want to ask?

RELATIONAL DATABASE

Rows and columns
`Paper(id, title, year)`
Which papers are from 2025?

VECTOR DATABASE

Embedding vectors
$p_1 \mapsto (0.2, 0.8, \ldots)$
Which papers are similar to this one?

GRAPH DATABASE

Nodes and edges
$p_1 \xrightarrow{\text{cites}} p_2$
Which papers cite this paper?

Different ways to organize and query data; all typically encoded as **bits**.

Data Model vs. Information Representation

DATA MODEL: HOW DO WE ORGANIZE INFORMATION?

Tables, vectors, and graphs describe the structure visible to applications.

INFORMATION REPRESENTATION: WHAT CARRIES THE INFORMATION?

Classical bit
A definite 0 or 1. Text, identifiers, and vector coordinates are encoded using strings of bits.

Quantum bit (qubit)
A state can be $\alpha|0\rangle + \beta|1\rangle$. Measuring in this basis gives 0 or 1, not both values to read out.

A vector of numbers in a database is still classical data; it is not automatically a quantum state.

Reading: IBM Quantum Learning, Basics of Quantum Information: Single Systems.

What Do We Mean by Quantum Data?

Quantum data is information carried by quantum states. There are two useful starting points.

CLASSICAL DATA, QUANTUM PROCESSING

Start with ordinary records or candidate labels. Prepare a quantum encoding, apply an algorithm, then read a classical result.

DATA THAT IS ALREADY QUANTUM

Start with states produced by a quantum experiment or simulation. Measurement extracts information and can change the state.

In this lecture, we focus on the first route: can quantum processing help a database task? This changes the computation, not necessarily the logical data model.

Biamonte et al., Quantum Machine Learning, Nature 2017; IBM Quantum Learning: General Measurements.

Why Consider Quantum Databases?

Start with a basic database task: **find a record satisfying a condition**.

THE UNSTRUCTURED CASE: NO SHORTCUT TO THE ANSWER

Records have **no useful ordering, index, or other structure** that helps locate a match. We can only test candidates against the condition.

For one matching record among N candidates, classical search needs $\Theta(N)$ tests in the worst case. Grover search uses $O(\sqrt{N})$ oracle calls with high success probability.

This is a hard baseline for search: even when classical search cannot exploit structure, quantum computation can reduce the number of tests. Total runtime also depends on access and oracle costs.

Reading: Grover, A Fast Quantum Mechanical Algorithm for Database Search, STOC 1996.

Outline

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EVOLVE

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MOON

One Question about the Moon



When you are not looking at the moon, is it there?

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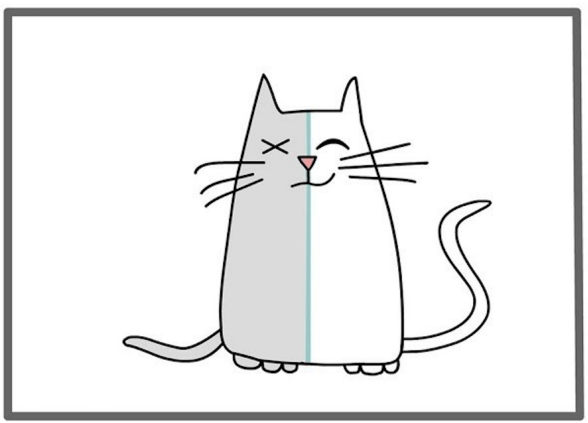
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CAT

Schrödinger's Cat



Electron: if we observe an electron, it will be in either of the two state: spin up, or spin down.

Schrödinger's Cat: There is a cat, a gun, and an electron, in a black box. If the electron spins down, the gun will automatically kill the cat; if the electron spins up, the gun will do nothing. If we open the box and observe, the cat will definitely either be alive or dead. The question is: if we don't open the box, is the poor cat alive or dead?

State

Classical state:

- Cat state: $|alive\rangle$ or $|dead\rangle$
- Bit state: $|0\rangle$ or $|1\rangle$
- Die state: $|1\rangle$, $|2\rangle$, $|3\rangle$, $|4\rangle$, $|5\rangle$, $|6\rangle$

Quantum state, when we are observing it:

- Electron state: $|up\rangle$ or $|down\rangle$
- Schrödinger's cat state: $|alive\rangle$ or $|dead\rangle$

What's the difference?

The difference shows up when we are not looking at the state.

Superposition: what is a state like when we are not observing it?

Classical state:

- Cat state: $|alive\rangle$ or $|dead\rangle$
- Bit state: $|0\rangle$ or $|1\rangle$
- Die state: $|1\rangle$, $|2\rangle$, $|3\rangle$, $|4\rangle$, $|5\rangle$, $|6\rangle$

Quantum state, when we are not observing it:

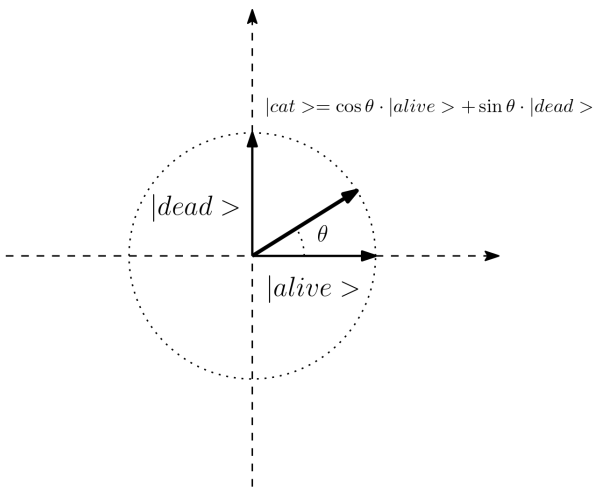
- Electron state: $|up\rangle + |down\rangle$
- Schrödinger's cat state: $|alive\rangle + |dead\rangle$

Note 1: Superposition state is a new state. It is neither $|alive\rangle$ nor $|dead\rangle$. It is $|alive\rangle + |dead\rangle$.

Note 2: The $+$ is vector addition. The displayed sums omit a normalization factor of $1/\sqrt{2}$.

Note 3: $|alive\rangle + |dead\rangle$ means that if we observe the cat, it will **randomly collapse** into either $|alive\rangle$ or $|dead\rangle$. But if we don't observe, it is neither of $|alive\rangle$ or $|dead\rangle$.

Bohr: God rolls a die.



$$|cat\rangle = \cos\theta \cdot |alive\rangle + \sin\theta \cdot |dead\rangle$$
$$P_{alive} = \cos^2\theta$$
$$P_{dead} = \sin^2\theta$$

State vector

For the real-amplitude example, we can rewrite $|cat\rangle$ using a vector form:

$$|alive\rangle \longrightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |dead\rangle \longrightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|cat\rangle = \cos\theta \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \sin\theta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}.$$

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EVOLVE

What is evolving?

Evolve: if a state $|A\rangle$ transits to a new state $|B\rangle$, we say it **evolves** from $|A\rangle$ to $|B\rangle$.

For example,

- if someone kills his cat, then $|cat\rangle$ will evolve from $|alive\rangle$ to $|dead\rangle$
- if I use a NOT gate to act on a bit $|0\rangle$, then $|0\rangle$ will evolve from $|0\rangle$ to $|1\rangle$
- If I push a stone, then $|stone\rangle$ will evolve from $|static\rangle$ to $|moving\rangle$

Mechanics: if we know a state, and we also know how it evolves, then we know **mechanics** of it.

$$\text{State} + \text{Evolving Rules} = \text{Mechanics}$$

For example: cat killer mechanics,

information mechanics, and,

Newton/Classical Mechanics

Have you notice the Newton mechanics?

In Newton/classical mechanics,

- the state of an object is a triplet: $|position, velocity, mass\rangle$,
- the evolving rule is Newton's second law: $F = ma$

$$|position, velocity, mass\rangle + \text{Newton's second law} = \text{Newton Mechanics}$$

For example, if we know the triplet state of a GPS satellite, we will definitely predict the state of it after one hour. This is because how a satellite state evolves follows the Newton's second law, and scientists know this evolving rule very well.

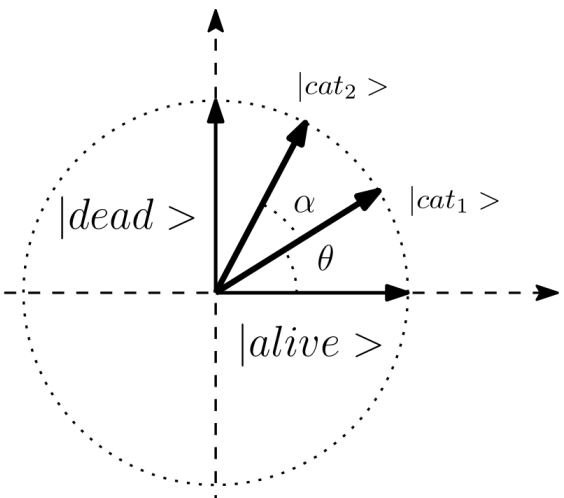
How a quantum Schrödinger's cat state evolves?

Quantum state $|cat\rangle$ evolves according to Schrödinger's equation:

$$-i\hbar\frac{\partial}{\partial t}|cat\rangle = (\frac{\hbar^2}{2m}\bigtriangledown^2 - V)|cat\rangle$$

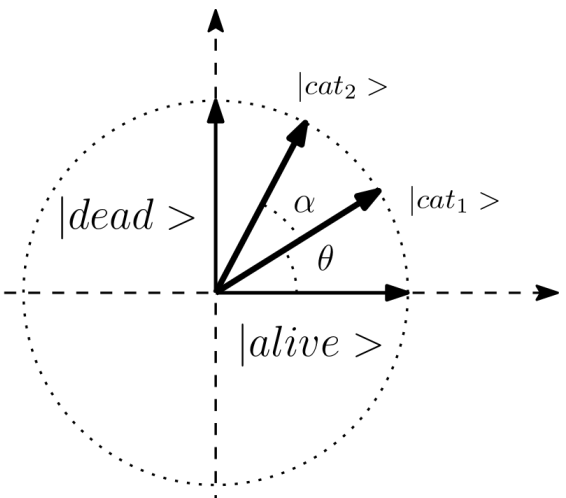
A quantum state evolves according to Schrödinger's equation, just like a classical triplet state evolves according to Newton second's law.

Quantum state evolves by vector rotation



- $|cat\rangle$ is a unit vector on the unit circle
- How to evolve a unit vector to another unit vector on the same unit circle?
- Yes, by **vector rotation**!
- How to do vector rotation?

Yes, by matrix multiplication!

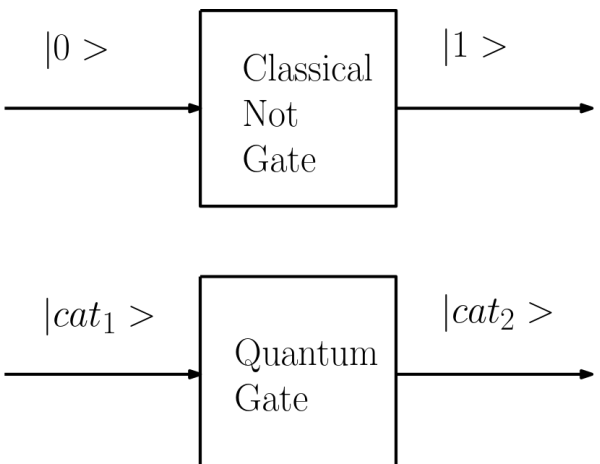


$$|cat_1\rangle = \begin{bmatrix}\cos\theta \\ \sin\theta\end{bmatrix}, |cat_2\rangle = \begin{bmatrix}\cos(\theta+\alpha) \\ \sin(\theta+\alpha)\end{bmatrix}$$

$$|cat_2\rangle = \begin{bmatrix}\cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha\end{bmatrix} \cdot |cat_1\rangle$$

The rotation matrix rotates a unit vector with angle θ to another unit vector with angle $\theta + \alpha$ counterclockwise.

Quantum Gate



For our two-state, real-amplitude example, the evolution over a chosen interval can be represented by a simple 2×2 rotation matrix.

$$-i\hbar\frac{\partial}{\partial t}\left|cat\right\rangle = (\frac{\hbar^2}{2m}\bigtriangledown^2 - V)\left|cat\right\rangle$$

$$\rightsquigarrow \quad \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

Now, what is quantum computation?

Just like we explain mechanics as:

$$\text{Mechanics} = \text{State} + \text{Evolve}$$

The quantum computation could be understand as:

$$\text{Quantum Computation} = \text{Qubit} + \text{Quantum Gate}$$

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APPLE

Delicious Apple Searching Problem

Delicious Apple Searching Problem: There are N apples that look same. Only one of them is a sweet delicious apple (a good apple), while all the remaining $N - 1$ apples taste very bitter (bad apples). In the worst case how many times you have to try to find the good apple?

Note:

- No parallel method is allowed, i.e., you can't ask your friends to taste apples simultaneously
- We count the times in the worst case (worst-case time complexity)
- Unstructured database searching

One minute to think about this problem.

Classical Answer

Classical search requires $\Theta(N)$ **tests** in the worst case to find the good apple.

Quantum Algorithm: Grover's Search

States: $|good\rangle, |bad_1\rangle, |bad_2\rangle, \ldots, |bad_{N-1}\rangle$.

EQUAL SUPERPOSITION STATE

$$|apple\rangle = \frac{1}{\sqrt{N}}\Big(|good\rangle + |bad_1\rangle + \cdots + |bad_{N-1}\rangle\Big)$$

$$= \frac{1}{\sqrt{N}}|good\rangle + \frac{1}{\sqrt{N}}\sum_{i=1}^{N-1}|bad_i\rangle$$

Every apple has the same amplitude $1/\sqrt{N}$, hence the same measurement probability $1/N$.

Quantum Algorithm: Grover's Search

Now define two normalized vectors, exactly as before:

$$|GOOD\rangle = |good\rangle\,, \qquad |BAD\rangle = \frac{1}{\sqrt{N-1}} \sum_{i=1}^{N-1} |bad_i\rangle$$

Thus $\sum_{i=1}^{N-1} |bad_i\rangle = \sqrt{N-1}\,|BAD\rangle$. Substitute this into the previous expression:

$$|apple\rangle = \frac{1}{\sqrt{N}}\,|GOOD\rangle + \sqrt{\frac{N-1}{N}}\,|BAD\rangle$$

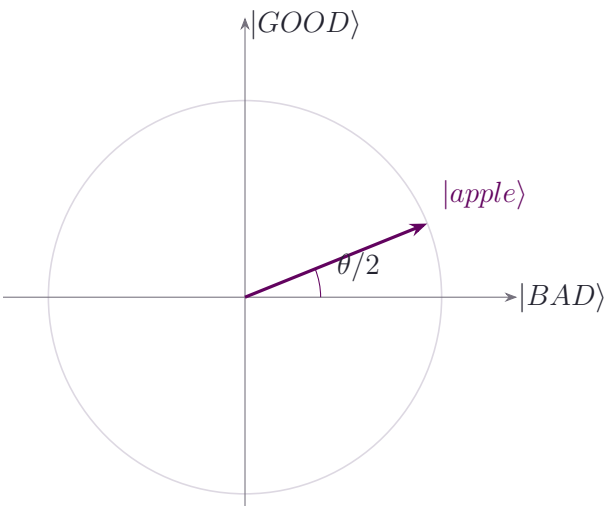
GOOD represents the one delicious apple; BAD combines all the bitter apples into one direction.

Quantum Algorithm: Grover's Search

$$\cos\frac{\theta}{2} = \sqrt{\frac{N-1}{N}}, \qquad \sin\frac{\theta}{2} = \frac{1}{\sqrt{N}}$$

$$|apple\rangle = \cos\frac{\theta}{2}|BAD\rangle + \sin\frac{\theta}{2}|GOOD\rangle$$

For large N, $\theta/2$ is small. The initial vector is close to BAD, so an immediate observation is very likely to give a bitter apple.



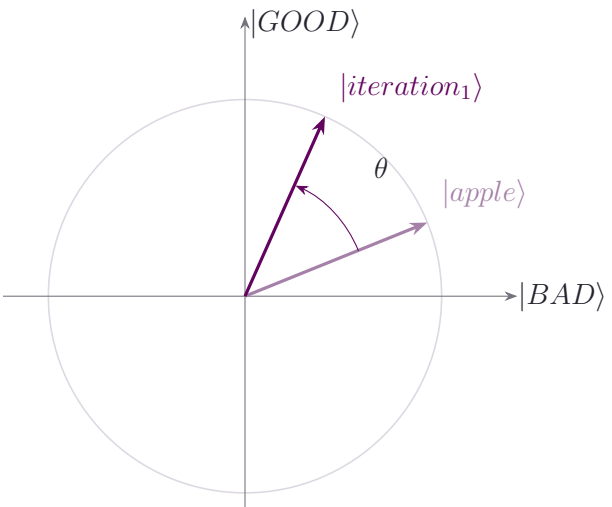
Quantum Algorithm: Grover's Search

How can we evolve $|apple\rangle$ toward $|GOOD\rangle$? Yes, by rotation.

One Grover iteration rotates the vector counterclockwise by θ:

$$\frac{\theta}{2} \quad \longrightarrow \quad \frac{\theta}{2} + \theta = \frac{3\theta}{2}$$

Repeat the same rotation. Stop when the vector is close to GOOD; then observe it.



Quantum Algorithm: Grover's Search

The initial angle is $\theta/2$. Each iteration adds θ.

$$|apple\rangle = \cos\frac{\theta}{2}|BAD\rangle + \sin\frac{\theta}{2}|GOOD\rangle$$

$$|iteration_1\rangle = \cos\frac{3\theta}{2}|BAD\rangle + \sin\frac{3\theta}{2}|GOOD\rangle$$

$$|iteration_k\rangle = \cos\frac{(2k+1)\theta}{2}|BAD\rangle + \sin\frac{(2k+1)\theta}{2}|GOOD\rangle$$

After k rotations: $\theta/2 + k\theta = (2k+1)\theta/2$. The GOOD probability is the square of its coefficient.

Quantum Algorithm: Grover's Search

We want the final vector to be close to GOOD, whose angle is $\pi/2$.

$$\frac{(2k+1)\theta}{2} \approx \frac{\pi}{2}$$

$$2k+1 \approx \frac{\pi}{\theta} \qquad \Longrightarrow \qquad k \approx \frac{\pi}{2\theta} - \frac{1}{2}$$

Choose a nearby integer k. The final angle is usually close to, rather than exactly, $\pi/2$.

Quantum Algorithm: Grover's Search

From the initial state, $\sin(\theta/2) = 1/\sqrt{N}$. For large N, use $\sin u \approx u$ (in radians):

$$\frac{\theta}{2} \approx \frac{1}{\sqrt{N}} \qquad \Longrightarrow \qquad \theta \approx \frac{2}{\sqrt{N}}$$

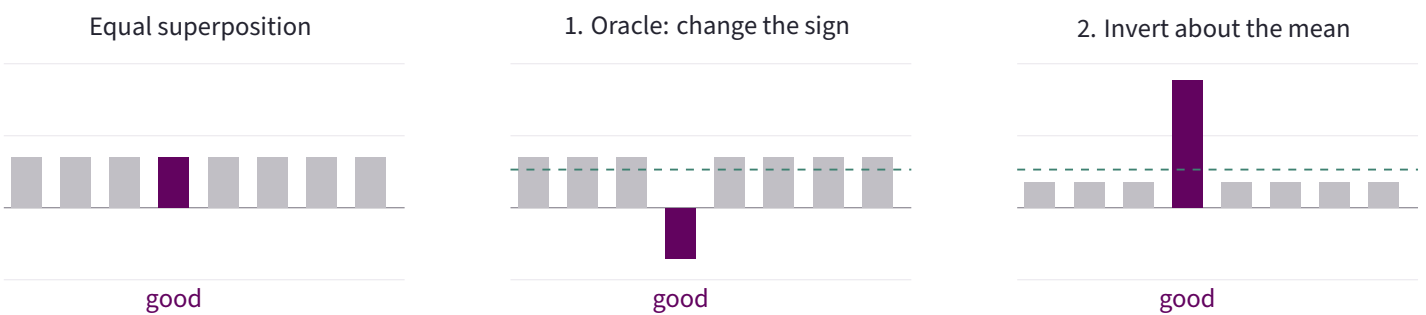
$$k \approx \frac{\pi}{2\theta} - \frac{1}{2} \approx \frac{\pi}{4}\sqrt{N} - \frac{1}{2} = O(\sqrt{N})$$

After $O(\sqrt{N})$ rotations, observe the state: it gives the delicious apple with high probability. This counts search iterations (each using one test), not data-loading cost.

Grover Search: An Alternative Derivation

An alternative derivation through two amplitude flips. Eight states are listed in the same order as the bars.

$$|apple\rangle = \frac{1}{\sqrt{8}}\left(|bad_1\rangle + |bad_2\rangle + |bad_3\rangle + |good\rangle + |bad_4\rangle + \cdots + |bad_7\rangle\right)$$



Purple marks the good apple. The dashed line is the mean after marking; the second flip reflects every amplitude about this line.

Step 1: Mark the Good Apple

Start with eight equal amplitudes, each $a = 1/\sqrt{8}$.

ORACLE: CHANGE THE SIGN

$$a_{good} : a \longmapsto -a$$

$$a_{bad} : a \longmapsto a$$

MEAN AFTER MARKING

$$\mu = \frac{7a - a}{8} = \frac{3}{4}a$$

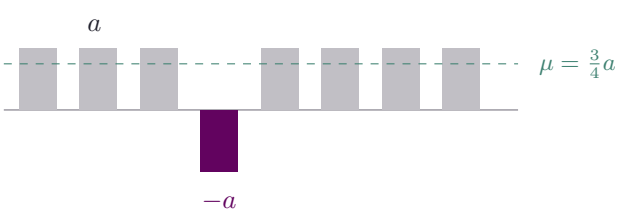
Seven positive amplitudes and one negative amplitude are averaged.

Marking alone does not change probabilities: $|-a|^2 = |a|^2 = 1/8$. It prepares the next step.

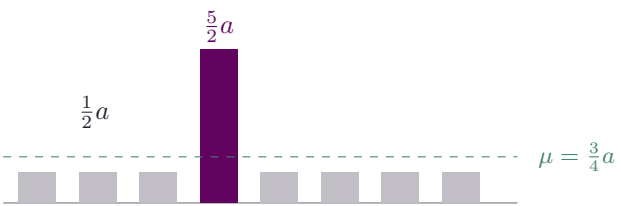
Step 2: Invert About the Mean

$a = 1/\sqrt{8}$, $\mu = 3a/4$. Reflect each amplitude: $a_i \mapsto 2\mu - a_i$.

BEFORE: AFTER THE ORACLE



AFTER: REFLECT ABOUT THE MEAN



Good: $2\mu - (-a) = \frac{5}{2}a$

Each bad: $2\mu - a = \frac{1}{2}a$

Good probability: $25/32 = 0.78125$; each bad: $1/32 = 0.03125$. Total: $25/32 + 7(1/32) = 1$.

From Two Flips to a Recurrence

One good apple: g_k is its amplitude; b_k is the amplitude of each bad apple. Initially, $g_0 = b_0 = 1/\sqrt{N}$. Let $h_k = \sqrt{N-1}\, b_k$ to simplify the algebra ($N > 1$).

1. SUBSTITUTE THE MEAN

$$\mu_k = \frac{(N-1)b_k - g_k}{N}$$

$$g_{k+1} = 2\mu_k + g_k$$

$$= \left(1 - \frac{2}{N}\right) g_k + \frac{2(N-1)}{N} b_k$$

$$b_{k+1} = -\frac{2}{N} g_k + \left(1 - \frac{2}{N}\right) b_k$$

2. REWRITE THE COEFFICIENTS

$$\sin\alpha = \frac{1}{\sqrt{N}}, \quad \cos\alpha = \frac{\sqrt{N-1}}{\sqrt{N}}$$

$$\cos 2\alpha = 1 - \frac{2}{N}, \quad \sin 2\alpha = \frac{2\sqrt{N-1}}{N}$$

$$g_{k+1} = \cos 2\alpha\, g_k + \sin 2\alpha\, h_k$$

$$h_{k+1} = \cos 2\alpha\, h_k - \sin 2\alpha\, g_k$$

Solve the Recurrence: The First Round

Recall: g_k is the good amplitude; $h_k = \sqrt{N-1}\, b_k$ combines the equal bad amplitudes. Choose $\alpha = \arcsin(1/\sqrt{N})$, so $g_0 = \sin\alpha$ and $h_0 = \cos\alpha$.

START WITH THE RECURRENCES FROM THE PREVIOUS PAGE

$$g_{k+1} = \cos 2\alpha\, g_k + \sin 2\alpha\, h_k, \qquad h_{k+1} = \cos 2\alpha\, h_k - \sin 2\alpha\, g_k$$

SUBSTITUTE THE INITIAL VALUES ($k = 0$)

$$g_1 = \cos 2\alpha \sin\alpha + \sin 2\alpha \cos\alpha = \sin(2\alpha + \alpha) = \sin 3\alpha$$

$$h_1 = \cos 2\alpha \cos\alpha - \sin 2\alpha \sin\alpha = \cos(2\alpha + \alpha) = \cos 3\alpha$$

We used $\sin(u+v) = \sin u \cos v + \cos u \sin v$ and $\cos(u+v) = \cos u \cos v - \sin u \sin v$.

From One Round to k Rounds

The same addition identities apply at every round. Suppose $g_k = \sin\beta_k$ and $h_k = \cos\beta_k$.

$$g_{k+1} = \cos 2\alpha \sin\beta_k + \sin 2\alpha \cos\beta_k = \sin(\beta_k + 2\alpha)$$

$$h_{k+1} = \cos 2\alpha \cos\beta_k - \sin 2\alpha \sin\beta_k = \cos(\beta_k + 2\alpha)$$

THE ARGUMENT INCREASES BY 2α EACH TIME

$$\beta_0 = \alpha \longrightarrow \beta_1 = 3\alpha \longrightarrow \beta_2 = 5\alpha \longrightarrow \beta_k = (2k+1)\alpha$$

$$g_k = \sin((2k+1)\alpha), \qquad b_k = \frac{h_k}{\sqrt{N-1}} = \frac{\cos((2k+1)\alpha)}{\sqrt{N-1}}$$

When Should We Stop?

Measurement succeeds with the squared good amplitude: $P_k = g_k^2 = \sin^2((2k+1)\alpha)$.

1. AIM FOR THE FIRST PEAK: $\sin^2(\pi/2) = 1$

$$(2k+1)\alpha \approx \frac{\pi}{2} \quad \Rightarrow \quad k \approx \frac{\pi}{4\alpha} - \frac{1}{2}$$

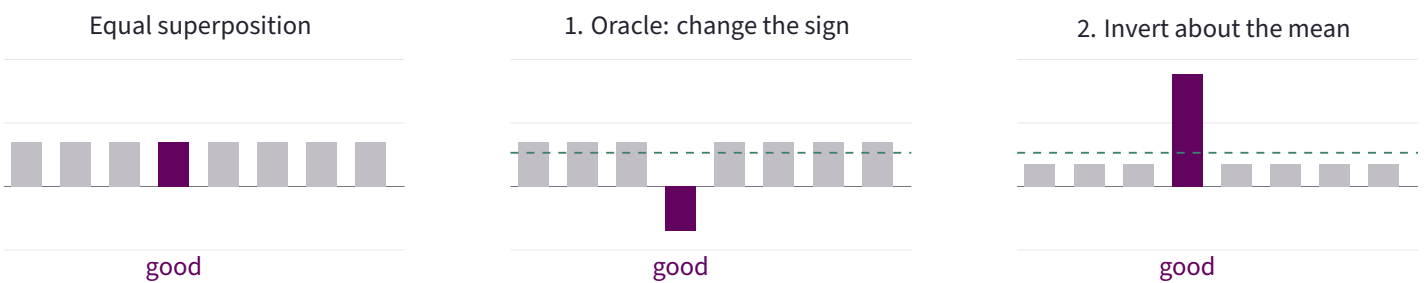
2. SUBSTITUTE $\alpha = \arcsin(1/\sqrt{N})$

$$k \approx \frac{\pi}{4\arcsin(1/\sqrt{N})} - \frac{1}{2} \approx \frac{\pi}{4}\sqrt{N} - \frac{1}{2} = O(\sqrt{N})$$

For large N, $\arcsin(1/\sqrt{N}) \approx 1/\sqrt{N}$. Choose the nearest integer k: the probability is close to one, not necessarily exactly one. More rounds can move past the peak.

The Oracle: Where the Problem Is Encoded

$$O_f |x\rangle = (-1)^{f(x)} |x\rangle , \qquad f(x) = 1 \text{ if } x \text{ is good}$$



The oracle tells the algorithm what counts as good. It encodes the problem condition and marks each satisfying component by changing its sign, as shown in the middle panel.

To apply Grover search to a new problem, design its oracle. The amplitude-amplification steps then increase the probability of measuring a marked answer.

Algorithm Notes

- Approximation algorithm? No: a bounded-error search algorithm.
  - With a suitable iteration count, the failure probability is at most $1/N$. A successful output is the exact good label.
- Parallel algorithm? It seems that the equal superposition state contains all of the possible states. When we use a rotation matrix to evolve the state, it is just like we are operating on all of the states **simultaneously**. This motivates different interpretations of the superposition: $|cat\rangle = |alive\rangle + |dead\rangle$
  - Opinion 1: superposition state is **one** state.
  - Opinion 2: superposition state contains **multiple** states.

These are interpretations; the calculation does not assume parallel universes.

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Annealing

From Search to Optimization

Grover searches for a marked answer. Annealing seeks a low-cost configuration.

Find x that minimizes $E(x)$

Think of $E(x)$ as an energy landscape: valleys are good solutions. We will compare classical simulated annealing with quantum annealing, then build a small database example.

Classical Simulated Annealing

Maintain one candidate x. Propose a nearby candidate x' (for example, flip one bit).

ACCEPT OR REJECT THE MOVE

$$\Delta E = E(x') - E(x)$$

Accept improvements ($\Delta E \leq 0$).
For a worse move, accept with probability $e^{-\Delta E/T}$.

LOWER THE TEMPERATURE T

High T: uphill moves are more likely, helping escape local minima.

Low T: improvements dominate. Repeat moves as T decreases.

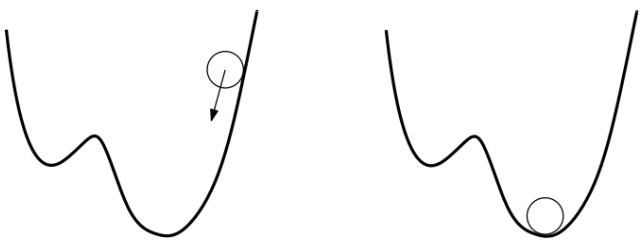
Example: an uphill move with $\Delta E = 2$ is accepted with probability $e^{-1} \approx 0.37$ at $T = 2$, but $e^{-4} \approx 0.018$ at $T = 0.5$.

Reference: D-Wave simulated annealing sampler (Metropolis acceptance).

Two Ways to Find a Low-Energy State

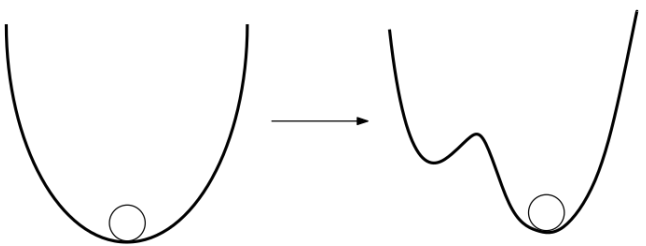
The landscape represents objective value (energy): lower valleys correspond to better solutions.

CLASSICAL SIMULATED ANNEALING



Explore a fixed landscape. Thermal uphill moves help escape local minima. Lower T to favor low-energy states.

QUANTUM ANNEALING



Gradually change the landscape. Start from an easy ground state; reduce the driver as the problem energy takes over.

Quantum transitions can include tunneling through barriers; the right-hand sketch shows the changing landscape.

How Quantum Annealing Differs

Encode the objective as an energy function of a quantum system. Gradually change the system from an easy initial state toward the problem energy landscape.

$$H(s) = A(s)H_{\text{driver}} + B(s)H_{\text{problem}}, \qquad 0 \leq s \leq 1$$

H describes energy. Reduce the driver strength $A(s)$ and increase the problem strength $B(s)$. The driver enables quantum transitions, including tunneling through barriers.

SIMULATED ANNEALING

Classical random moves; temperature controls uphill acceptance.

QUANTUM ANNEALING

Quantum evolution; measure candidate bitstrings at the end and repeat.

Finite runs need not find the optimum. Neither method guarantees a speed advantage on every problem.

Reference: D-Wave, What is Quantum Annealing?

QUBO: A Common Optimization Form

Quadratic Unconstrained Binary Optimization

$$\min_{x_i \in \{0,1\}} E(x) = \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j + c$$

BINARY

x_i is a yes/no decision.
Example: build index i?

QUADRATIC

Linear terms and pairwise products; $x_i^2 = x_i$.

UNCONSTRAINED

Constraints are absorbed into penalty terms.

QUBO is a model, not an algorithm. Classical solvers can use it too. A quantum annealer uses an equivalent Ising form, via $s_i = 2x_i - 1$.

Reference: D-Wave, QUBOs and Ising Models.

Example: Choose Exactly One Index

Which index should we build to obtain the larger performance benefit?

THE PROBLEM

For a fixed query workload, two candidate indexes are available. Their estimated reductions in total runtime are:

Index A: **3 ms saved**
Index B: **2 ms saved**

Requirement: build exactly one. Building neither or both is invalid. Maximize the time saved.

1. WRITE THE BENEFIT TERM

$x_A, x_B \in \{0, 1\}$: 1 means build that index; 0 means do not build it.

$$B(x_A, x_B) = 3x_A + 2x_B$$

QUBO **minimizes**, so use $-B = -3x_A - 2x_B$. Larger benefit gives a smaller objective value.

Benefit alone favors choosing both: $-B(1, 1) = -5$. We must add a penalty to enforce “exactly one.”

Turn the Constraint into a Penalty

The constraint is $x_A + x_B = 1$. Move everything to the left, square the violation, and multiply by a positive weight λ: $P = \lambda(x_A + x_B - 1)^2$.

2. CHECK WHAT THE SQUARE DOES

Choice	$x_A + x_B$	Violation	P
Neither	0	-1	λ
Exactly one	1	0	0
Both	2	1	λ

$$E = \underbrace{-3x_A - 2x_B}_{\text{negative benefit}} + \underbrace{4(x_A + x_B - 1)^2}_{\text{constraint penalty}}$$

3. CHOOSE ENOUGH PENALTY

Both indexes give energy $-5 + \lambda$. A alone gives -3. To make both worse, require $-5 + \lambda > -3$, i.e. $\lambda > 2$. Choose $\lambda = 4$.

Since $x_i^2 = x_i$, this is the QUBO $E = 4 - 7x_A - 6x_B + 8x_Ax_B$. Next: verify every choice.

Check the QUBO and Read the Answer

Evaluate all four choices: $E = -3x_A - 2x_B + 4(x_A + x_B - 1)^2$. Lower is better.

x_A	x_B	Benefit	Penalty	Energy
0	0	0	4	4
1	0	3	0	$\mathbf{-3}$ (minimum)
0	1	2	0	-2
1	1	5	4	-1

RESULT: BUILD INDEX A

The penalty prevents choosing both. Here weight 4 is sufficient: both scores -1, worse than the feasible optimum -3.

A solver returns candidate bitstrings. Decode each one, check feasibility, and compare energies.

This table proves the toy optimum. Real annealing runs sample candidates; they need not return the optimum every time.

Outline

MOON

CAT

EVOLVE

APPLE

Annealing

END

END

When you are not looking at the moon, is it there?



We don’t know